DATA SCIENCE WITH PYTHON LOG SHEET – 60 DAYS

S.NO	DATE	ACTIVITIES / CONCEPTS COVERED	SIGNATURE
1		Introduction to Data Science – Discussed fundamentals and applications	
2		Python Installation & Setup – Configured Python environment	
3		Python IDEs – Explored coding platforms	
4		Python Variables & Data Types – Practiced variables and data types	
5		Control Flow Statements – Implemented conditional logic	
6		Loops in Python – Worked with loops	
7		Python Functions – Created reusable functions	
8		Exception Handling – Handled errors using try-except	
9		Python Data Structures – Worked with Python collections	
10		NumPy Basics – Performed numerical operations	
11		Pandas Basics – Created DataFrames and Series	
12		Loading Data from Files – Imported CSV and Excel files	
13		SQL Data Loading – Connected SQL databases	
14		MongoDB Integration – Integrated MongoDB with Pandas	
15		Web Scraping – Extracted data from websites	
16		Matplotlib Visualization – Built charts and graphs	
17		Seaborn Visualization – Created statistical plots	
18		Plotly Visualization – Designed interactive dashboards	
19		Project – I Planning – Planned project workflow	
20		Data Preprocessing – Cleaned and prepared data	
21		Handling Missing Data – Handled missing values	
22		Data Aggregation & Grouping – Performed grouping and aggregation	
23		Data Transformation – Applied normalization techniques	
24		Feature Engineering – Generated new features	

25		Encoding Categorical Data – Applied encoding methods	
26		Handling Outliers – Detected and treated outliers	
27		Handling Imbalanced Data – Balanced imbalanced datasets	
28		Exploratory Data Analysis – Performed exploratory analysis	
29		Hypothesis Testing & Statistical Tests – Conducted statistical testing	
30		Final Visualization & Project Completion – Presented project findings	
31		Introduction to Machine Learning – Learned ML concepts	
32		Supervised Learning – Worked with labeled datasets	
33		Unsupervised Learning – Explored clustering methods	
34		Regression Algorithms – Implemented regression models	
35		Classification Algorithms – Built classification models	
36		Decision Trees – Applied decision tree algorithms	
37		Random Forest – Used ensemble learning methods	
38		Support Vector Machines – Implemented SVM classifiers	
39		K-Nearest Neighbors – Worked with KNN algorithm	
40		Naive Bayes – Applied probabilistic learning	
41		Model Evaluation Metrics – Evaluated model performance	
42		Train-Test Split – Split data for training/testing	
43		Cross Validation – Performed model validation	
44		Hyperparameter Tuning – Optimized model parameters	
45		Clustering using K-Means – Applied K-Means clustering	
46		Hierarchical Clustering – Worked on hierarchical clustering	
47		Principal Component Analysis – Reduced dimensionality	
48		Time Series Analysis – Analyzed time-based data	
49		Natural Language Processing Basics – Processed textual datasets	
50		Text Preprocessing – Performed text cleaning	
51		Sentiment Analysis – Built sentiment prediction models	

52		Deep Learning Introduction – Explored deep learning basics	
53		Artificial Neural Networks – Built ANN models	
54		TensorFlow Basics – Worked with TensorFlow framework	
55		Keras Model Building – Designed models using Keras	
56		CNN Basics – Implemented convolutional networks	
57		RNN Basics – Worked with recurrent networks	
58		Mini Project Development – Developed mini project	
59		Project Testing & Validation – Tested and validated project	
60		Final Project Presentation – Presented final project	